

Introduction to Modern Artificial Intelligence

Artificial Intelligence (AI) is a field of computer science focused on building systems that can perform tasks requiring human intelligence. These tasks include image recognition, language understanding, reasoning, planning, and decision making.

1. Machine Learning

Machine Learning (ML) is a subset of AI where algorithms learn patterns from data. Instead of being explicitly programmed for every task, ML systems improve their performance through experience. Common categories include supervised learning, unsupervised learning, and reinforcement learning.

2. Neural Networks

Neural networks are computational models inspired by the structure of the human brain. They consist of interconnected layers of artificial neurons that process information. Deep learning models are large neural networks capable of solving complex tasks such as speech recognition and computer vision.

3. Applications of AI

Field	Example Applications
Healthcare	Medical image analysis, disease prediction
Finance	Fraud detection, algorithmic trading
Education	Personalized tutoring systems
Transportation	Self-driving vehicles and route optimization

4. Challenges and Ethics

Despite its benefits, AI raises important ethical concerns. Researchers and governments are discussing topics such as privacy, algorithmic bias, transparency, and the impact of automation on employment. Responsible AI development aims to ensure fairness, accountability, and safety.

AI continues to evolve rapidly and is becoming integrated into everyday life. Understanding its foundations and limitations is increasingly important for students, engineers, and society as a whole.